

# PURE optimization

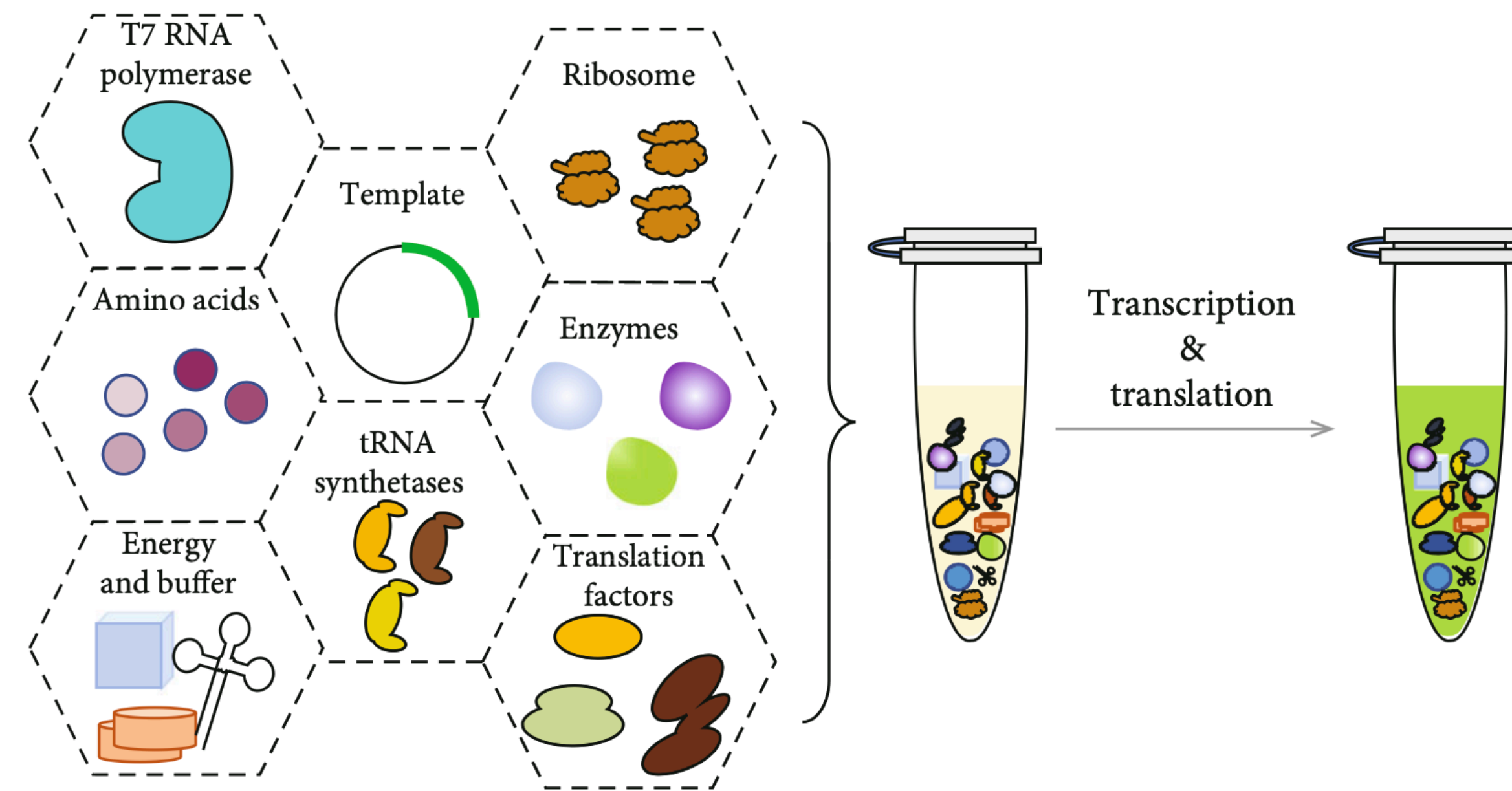
b.next <> FindWhatMatters.ai

June 2025

# What is PURE?

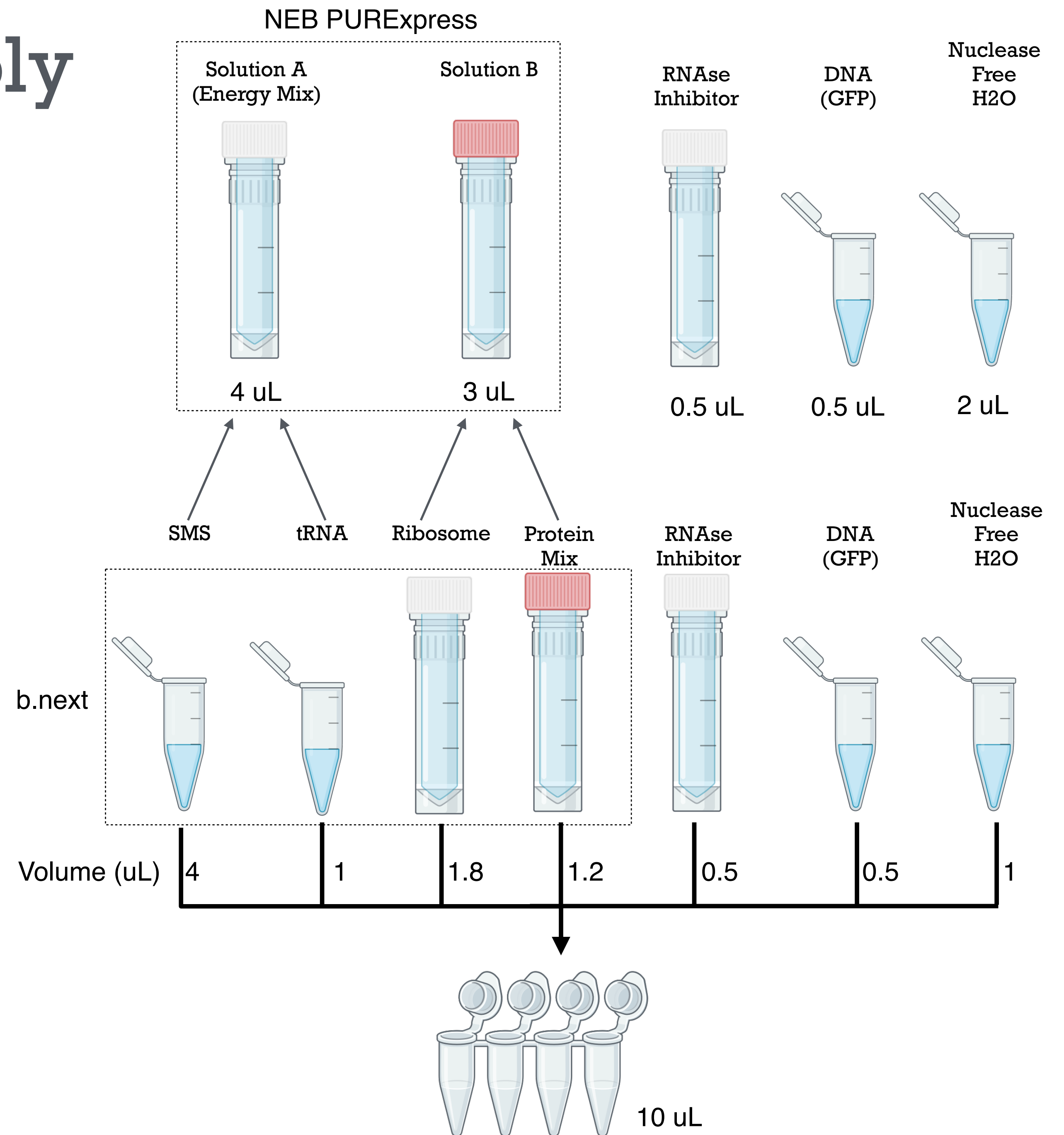
## (Protein synthesis Using Recombinant Elements)

- A cell-free protein synthesis system reconstituted from individually purified components required for **transcription** and **translation**.
- 48 components in PURE system for transcription and translation.
- Limitations:
  - Costly - due to the need for purified components
  - Lower yield compared to crude extract-based systems.
- Goal: Increase yield of PURE by finding the optima of each component with FindWhatMatters AI



# PURE reaction assembly

- Difficult to add each component individually → Grouping of components into mixtures
- NEB PURExpress is grouped into 2 solutions: A & B
  - NEB Sol A = Small molecules solution + tRNAs
  - NEB Sol B = Ribosomes & protein mix
- At bnext, we group into 4 components in addition to RNase inhibitor, DNA, and nuclease free water.
  - Keep **volume constant** but change component concentration within mixture.





# Division of Components based on Function

| Included In              | Categorization          | Component                                                     | Function                                                         | Used In                      | Physical Type        |
|--------------------------|-------------------------|---------------------------------------------------------------|------------------------------------------------------------------|------------------------------|----------------------|
| Ribosome                 | Ribosomal Machinery     | 70S Ribosome                                                  | mRNA translation → polypeptide                                   | Translation                  | Ribonucleoprotein    |
| Protein Mix              | Transcription Machinery | T7 RNA Polymerase                                             | DNA → mRNA synthesis via T7 promoter                             | Transcription                | Protein (enzyme)     |
|                          | Translation Factors     | IF1, IF2, IF3                                                 | Initiate translation                                             | Translation                  | Proteins             |
|                          |                         | EF-Tu, EF-Ts, EF-G                                            | Deliver tRNA, promote translocation                              | Translation                  | Proteins             |
|                          |                         | RF1, RF2, RF3                                                 | Recognize stop codons                                            | Translation                  | Proteins             |
|                          |                         | RRF                                                           | Recycles ribosome after termination                              | Translation                  | Protein              |
|                          | Aminoacylation          | 20 Aminoacyl-tRNA Synthetases (-RS)                           | Attach AA to tRNA (uses ATP → AMP + PPi)                         | tRNA Charging                | Proteins             |
|                          | Pyrophosphatase         | Inorganic PPase                                               | Hydrolyzes PPi (from transcription and aminoacylation)           | Transcription, tRNA charging | Protein (enzyme)     |
|                          | Energy Regeneration     | Creatine Kinase                                               | ATP regeneration                                                 | Both                         | Protein (enzyme)     |
|                          |                         | Creatine Phosphate                                            | Phosphagen energy donor                                          | Both                         | Small molecule       |
| Small Molecules Solution | Nucleotides             | ATP, GTP                                                      | ATP: transcription & tRNA charging; GTP: translation             | Both                         | Nucleotides          |
|                          |                         | CTP, UTP                                                      | CTP & UTP for mRNA synthesis                                     | Transcription                | Nucleotides          |
|                          | Amino Acids             | 20 Standard AAs                                               | Protein building blocks                                          | Translation                  | Small molecules      |
|                          | Buffers & Ions          | HEPES                                                         | pH buffering (~7.5–8.0)                                          | Both                         | Buffer               |
|                          |                         | Mg <sup>2+</sup> (MgCl <sub>2</sub> or Mg(OAc) <sub>2</sub> ) | Essential for ribosome, tRNA synthetase, and polymerase function | Both                         | Salt/Ion             |
|                          |                         | K <sup>+</sup> (K-glutamate or KCl)                           | Mimics cytosolic ionic environment                               | Both                         | Salt/Ion             |
|                          |                         | Na <sup>+</sup> (optional, NaCl)                              | Stabilizes tRNA folding and ribosomes (sometimes omitted)        | Translation                  | Salt/Ion (optional)  |
|                          | Small Molecules         | TCEP                                                          | Maintains reducing environment, prevents protein oxidation       | Both                         | Reducing agent       |
|                          |                         | Spermidine                                                    | Stabilizes nucleic acids and ribosomes                           | Both                         | Polyamine (cationic) |
| tRNA                     | tRNAs                   | Total tRNA mix (E. coli)                                      | Provide codon-matching amino acid delivery                       | Translation                  | RNA                  |



# Restrictions

- PURE reaction volumes constant
  - For our initial runs, we will limit the volume of each mixture to the volumes in table.
  - Constant concentrations & volumes: RNase Inhibitor & Water/ headroom
- Protein Mixtures:
  - No protein should be 0.
  - The overall protein concentration should be 8 mg/mL - 25 mg/mL (0.96 - 3 mg/mL final PURE reaction concentrations).
- Small Molecule Mixtures:
  - The volume of the energy mixture cannot go over 200 uL, meaning the capable concentrations of components are limited to volume.
    - Ex: HEPES concentration is currently 50 mM in PURE reaction (adding 25 uL of 1M stock = 77 uL headroom/ water). We can only increase concentration to 200 mM because we will be increasing volume of HEPES to 100 uL, leaving only 2 uL of headroom/ volume.

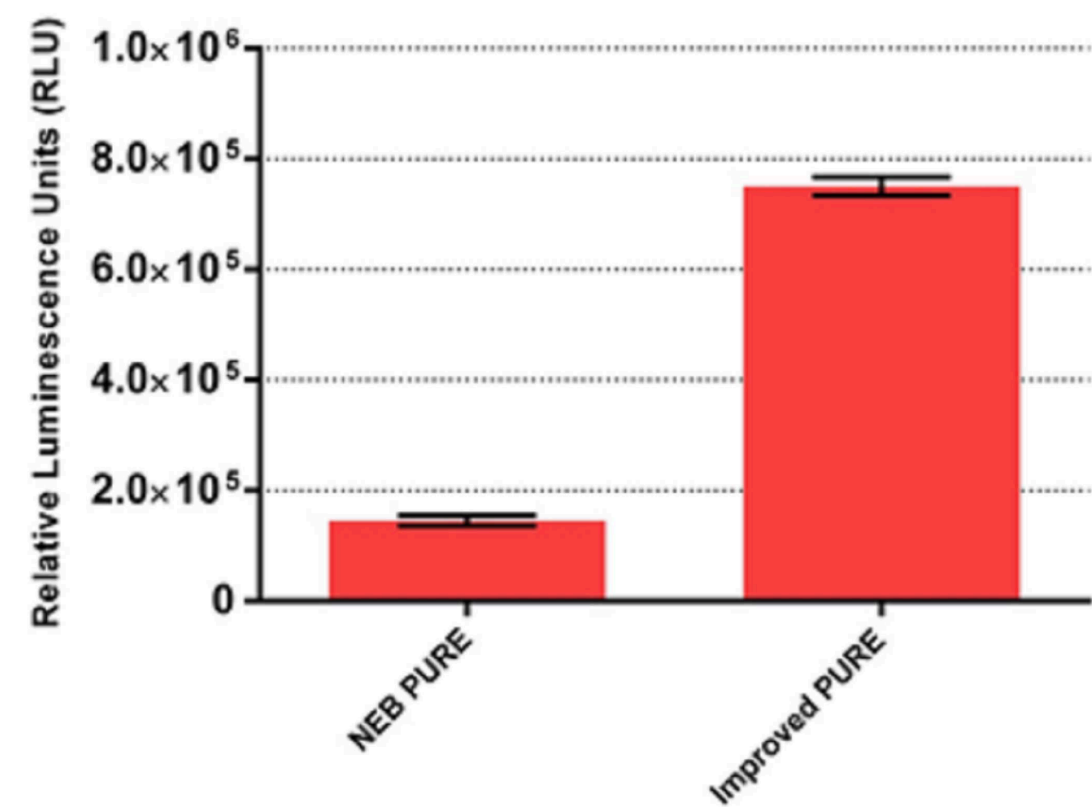
|                          | Volume (uL) |
|--------------------------|-------------|
| Protein Mix              | 1.2         |
| Ribosome                 | 1.8         |
| Small Molecules Solution | 4           |
| tRNA                     | 1           |
| DNA                      | 0.5         |
| Rnase Inhibitor          | 0.5         |
| Water/ headroom          | 1           |
| Total Volume (uL)        | 10          |

| Component            | Stock concentration [mM] | Concentration of components in reaction [mM] | Concentration in Energy solution [mM] | Final volume to add [uL] |
|----------------------|--------------------------|----------------------------------------------|---------------------------------------|--------------------------|
| HEPES                | 1000                     | 50 → 200                                     | 125                                   | 25.0 → 100               |
| Potassium glutamate  | 2500                     | 100                                          | 250                                   | 20.0                     |
| Magnesium acetate    | 1000                     | 7.5                                          | 18.75                                 | 3.8                      |
| ATP                  | 100                      | 2                                            | 5                                     | 10.0                     |
| GTP                  | 100                      | 2                                            | 5                                     |                          |
| CTP                  | 50                       | 1                                            | 2.5                                   |                          |
| UTP                  | 50                       | 1                                            | 2.5                                   |                          |
| Creatine phosphate   | 1000                     | 20                                           | 50                                    | 10.0                     |
| TCEP                 | 500                      | 1                                            | 2.5                                   | 1.0                      |
| Folinic acid         | 5                        | 0.02                                         | 0.05                                  | 2.00                     |
| Spermidine           | 200                      | 2                                            | 5                                     | 5.0                      |
| Amino Acid solution  | 3.25                     | 0.3                                          | 0.75                                  | 46.2                     |
| tRNA                 | 33.5                     |                                              |                                       |                          |
| Water                |                          |                                              |                                       | 77.1 → 2                 |
| Outer solution total |                          | Final concentration [fold]                   |                                       | Final volume             |
|                      |                          | 2.5                                          |                                       | 200                      |

# Previous publications on increasing PURE activity & yield

Goal: Increase protein expression of PURE by at least 5x

| Modification / Component Added                                                                                         | Function                                     | Quantitative Outcome                                                       |
|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------|
| EF-Tu alone                                                                                                            | Delivers aminoacyl-tRNAs to ribosome         | ~1.8× increase in Firefly luciferase (Fluc) yield                          |
| EF-Tu + EF-Ts + EF-G                                                                                                   | Full elongation/translocation support        | ~2.5× increase in Fluc yield                                               |
| RRF (Ribosome Recycling Factor)                                                                                        | Disassembles post-termination complex        | ~1.2× increase in Fluc synthesis                                           |
| IF1 + IF2 + IF3 (Initiation Factors)                                                                                   | Facilitate initiation complex formation      | Minor improvement, <b>not statistically significant</b>                    |
| All above translation factors combined                                                                                 | Comprehensive translation enhancement        | ~5× <b>increase</b> in Fluc yield vs. baseline PURE system                 |
| Custom PURE buffer ( <i>Key changes: 50 mM HEPES, 250 mM K-Glu, 10 mM Mg-Ac, 1.5 mM spermidine, 1 mM DTT, 2% PEG</i> ) | Optimized buffer composition for translation | ~2× <b>increase</b> in Fluc yield compared to standard manufacturer buffer |



Paper link: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0106232>